

Corruption Strategies for WAVE Training.

To train the WAVE model to distinguish between well-aligned and poorly aligned speech-text pairs, we implement five distinct corruption strategies that systematically introduce misalignments in Dutch and Portuguese transcripts. Each strategy simulates different types of synthesis errors commonly found in TTS-generated speech, while calculating precise correctness scores that indicate the expected proportion of correctly aligned words.

DUTCH EXPERIMENT:

Replace Strategy

Mechanism: Randomly selects one word in the transcript and replaces it with a word from the Dutch vocabulary set.

Vocabulary Used ; *ja, nee, en, de, het, een, van, is, dat, ik, je, we, maar, ook, nog, wel, niet, zijn, was, hebben, komen, gaan, maken, doen, kunnen, willen, moeten, zou, al, meer, zeer, waar, hier, daar, nu, dan, dus, want, omdat, zoals, misschien, natuurlijk, eigenlijk, die, deze, dit, ben, hem, hen, haar, er, voor, door, toen, ton, kan, ken, man, men, twee, drie, vier, vijf, zes, zeven, acht, negen, tien, eh, ehm, nou, tja, hoor, zeg, om, op, in, uit, bij, met, naar, na, over, onder, tussen, tegen, zonder, tot, af, aan*

Correctness Score: $(\text{total_words} - 1) / \text{total_words}$

Shuffle Strategy

Mechanism: Randomly selects a contiguous segment of 2 or more words and shuffles their order within that segment.

Vocabulary Used: No external vocabulary - operates on existing words in the transcript only.

Correctness Score: $\text{correctly_positioned_words} / \text{total_words}$

Drop Strategy

Mechanism: Removes one randomly selected word from the transcript, causing all subsequent words to shift their alignment positions.

Vocabulary Used: No external vocabulary - removes existing words from the transcript.

Correctness Score: $\text{drop_index} / \text{original_length}$

Add Strategy

Mechanism: Inserts a word from the Dutch vocabulary set at any position in the transcript, shifting subsequent words and disrupting their temporal alignment.

Vocabulary Used: Same as from the replace strategy.

Correctness Score: $\text{insertion_index} / (\text{original_length} + 1)$

Partial Strategy

Mechanism: Randomly truncates the transcript to either the first half or second half, simulating incomplete synthesis or premature termination.

Vocabulary Used: No external vocabulary - operates on existing words in the transcript only.

Correctness Score: 0.5

Single-Word Transcript (EDGE CASE):

Mechanism: Automatically receive addition corruption.

Vocabulary used: Same as from the replace strategy

Correctnes score: 0.5

PORTUGUESE EXPERIMENT:

Replace Strategy

Mechanism: Randomly selects one word in the transcript and replaces it with a word from the Dutch vocabulary set.

Vocabulary Used ; *sim, não, e, o, de, um, uma, tua, qualquer, coisa, deveria, gostaria, imaginemos.*

Correctness Score: $(\text{total_words} - 1) / \text{total_words}$

Shuffle Strategy

Mechanism: Randomly selects a contiguous segment of 2 or more words and shuffles their order within that segment.

Vocabulary Used: No external vocabulary - operates on existing words in the transcript only.

Correctness Score: $\text{correctly_positioned_words} / \text{total_words}$

Drop Strategy

Mechanism: Removes one randomly selected word from the transcript, causing all subsequent words to shift their alignment positions.

Vocabulary Used: No external vocabulary - removes existing words from the transcript.

Correctness Score: $\text{drop_index} / \text{original_length}$

Add Strategy

Mechanism: Inserts a word from the Dutch vocabulary set at any position in the transcript, shifting subsequent words and disrupting their temporal alignment.

Vocabulary Used: *sim, não, e, o, de, um, uma*

Correctness Score: $\text{insertion_index} / (\text{original_length} + 1)$

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Mechanism: Randomly truncates the transcript to either the first half or second half, simulating incomplete synthesis or premature termination.

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